Capital Circuit - Monopoly Game

CSC301 Group 4 | console Monopoly in Java (Abu Dhabi edition)

AI Usage Log for user CS (Caron Sylvia)

Course	CSC301 Data Structures and Algorithms - Dr Zakaria Khiati, Abu Dhabi University
Group / member	Group 4 | CS (Caron Sylvia)
AI tool	Claude (model claude-opus-5-5) in the Claude Project "Monopoly-Game"; Prompts 01.19-01.21 in a separate Claude desktop session
Period	1 November 2026, 16:09-19:32 GST | 23 prompts
Source log	C:\Claude\Monopoly\AI-Usage-log.txt (prompts verbatim, AI actions, AI responses, reflection candidates)
Brief reference	Section 7 (AI use and the AI log, 20% of the grade); Sections 7.1-7.3

1. Summary

1.1 How AI was used

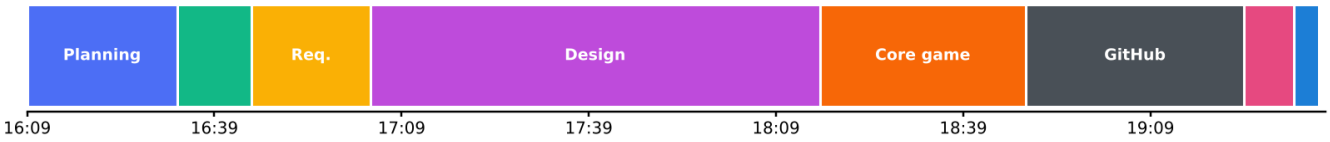
CS used Claude as the single AI tool of record for the group's Monopoly assignment, inside the Claude Project that holds the brief. Every instruction was logged verbatim in **AI-Usage-log.txt** together with what the AI did and how it answered, as Section 7 of the brief requires. The work moved in eight phases: planning, environment set-up, clarifying requirements, design, the core game, GitHub, enhancements and reporting.

The AI was given the brief's exact rules, constraints and method signatures rather than a general request, and work proceeded in small steps: no code was written until CS gave the go-ahead (Prompt 01.13). The AI produced the design (UML class diagram, board and card files), the Java source, the console board view and colour scheme, the build and test scripts, and the GitHub sync script. CS made the key decisions: deferring the twists until the assignment is confirmed, the bonus features, the visual console and player colours after play-testing, and keeping a real e-mail out of the public repository.

Every AI output was checked rather than trusted: 53 automated tests (including each Section 8.2 scenario reproduced with scripted dice), 1,000 random seeded games, and an automatic constraint check for allowed imports, banned collections and loop-free recursive methods. This surfaced five cases where the AI was wrong or incomplete (Section 2.3), each detected and fixed and recorded for the group's reflection.

23	8	25	53	1,000	5
prompts logged	work phases	Java source files	automated tests passing	seeded games checked	AI errors caught

Session phases, 1 November 2026 (GST)



Colours match the phase labels in the timeline (1.2). Time between prompts is counted in the phase of the preceding prompt; for example, the core build ran after the design prompt at 17:04.

1.2 Timeline of prompts and functionality built

Time (GST)	Prompt	Functionality built / outcome	Phase
16:09	**01.01**	AI log file created; brief analysed	Planning
16:12	**01.02**	Console UI prototype image	
16:23	**01.03**	Java toolchain check - JDK missing	
16:27	**01.04**	Build approach: Claude Project + VS Code	
16:33	**01.05**	One-click JDK 25 installer + setup skill	Environment
16:43	**01.06**	Environment verified; VS Code pinned to JDK 25	
16:45	**01.07**	Twist pool (T1-T8) explained	Requirements
16:47	**01.08**	Simplest twists shortlisted	
16:53	**01.09**	Twists deferred; bonus rules reviewed	
16:58	**01.10**	Language-toggle bonus assessed	
17:02	**01.11**	Bonus chosen: B4 visualiser + language	
17:04	**01.12**	UML class diagram, board and card files	Design
18:16	**01.13**	Core game: 24 classes, 44 tests, checker	Core game
18:35	**01.14**	Console board view (prototype look)	
18:45	**01.15**	Colour mode: player colours on squares	
18:49	**01.16**	GitHub sync script	GitHub
18:54	**01.17**	Push error 403 diagnosed; account switch	
18:57	**01.18**	Private no-reply e-mail for commits	
~19:02	**01.19**	Repo link checked (separate session)	
~19:07	**01.20**	Signed in with team account (separate session)	
19:12	**01.21**	Project pushed to CS-Monopoly branch	
19:24	**01.22**	Standings with properties; quit screen	Enhancements
19:32	**01.23**	This AI usage PDF report	Reporting

~ marks an approximate time: Prompts 01.19 and 01.20 were logged without a timestamp by a separate Claude session.

2. Detailed report

2.1 Log index (brief Section 7.1)

No.	Date	Member	Tool	What the conversation was for	Link / file name
01	01-11-2026	CS (Caron Sylvia)	Claude (claude-opus-5-5), Claude Project "Monopoly-Game"	Prompts 01.01-01.18 and 01.22-01.23: brief analysis, AI log, prototype, Java toolchain, environment set-up, twists and bonus, design, core game, board view, colour mode, GitHub sync script, standings, this report	session_01Rc15ip9Mj11V2i3BSohRjQ; shared link / PDF export: **to be added**
02	01-11-2026	CS (Caron Sylvia)	Claude, separate desktop-app session	Prompts 01.19-01.21: check the GitHub link, sign in with the team account, push the project to branch CS-Monopoly	Shared link / PDF export: **to be added**

The text log numbers all prompts under Conversation 01. Prompts 01.19-01.21 came from a separate Claude session, so they are listed here as conversation 02: Section 7.1 asks for one row and one unedited export per conversation. Both export links must be added before submission.

2.2 Prompt-by-prompt log

Prompts are quoted verbatim from AI-Usage-log.txt (spelling unchanged). The outcome column summarises the AI actions and response recorded there; the full text stays in the log file.

No.	Time (GST)	Phase	Prompt by CS (verbatim)	AI output and outcome
01.01	16:09	**Planning**	How this monopoly game must be developed is uploaded to the context section as a word document. Refer to the instructions and start creating the AI Log file, including this prompt. let me know the file path of the AI log file. Dont start developing the code, until i tell so. just read the context and keep in youe memory.	Read the brief from the Project; created AI-Usage-log.txt with the three parts the brief requires (index, full log, reflection). No code, as instructed.
01.02	16:12	**Planning**	now show me a prototype image of how this monopoly game will look like. generate this image based on the context file and referring to good monopoly games online. log this also	Looked at open-source terminal Monopoly projects; designed an original 24-square Abu Dhabi board; rendered a console mock-up (Prototype_Console_Mockup.png). Flagged that its money figures were illustrative.
01.03	16:23	**Planning**	Now show me the software components of JAVA that is used to build this monopoly game? Check my computer and let me know if i have the required software to build and run this game.	Listed the Java components the game needs; scanned folder names on the PC and reported no JDK; recommended Temurin JDK 25 LTS. Missed a JDK 17 on drive F: (reflection candidate 1).
01.04	16:27	**Planning**	ok then advise me how to build this game project? can i continue in Claude project or should i switch to VScode with Java extensions? or should i follow a hybrid approach, where claude project will suggest me the required AI skills which i can use in VS code?	Advised a hybrid: this Claude Project as the only AI of record, VS Code for build and debug with its own AI switched off (brief 7.3); proposed a phased build plan.
01.05	16:33	**Environment**	ok i will continue to use this Claude Project only for this monopoly development. Pls install the required software on this computer. Call this section as software environment setup and this entire activity create as a skill file (give appropriate naming for this skill file) in the project folder.	Created a one-click installer (Setup-Environment.bat/.ps1), AI-off VS Code settings and the csc301-java-environment-setup skill. Could not run the installer itself (no terminal input, UAC).
01.06	16:43	**Environment**	i had run the setup\Setup-Environment.bat look like required java components are installed and paths are setup. Pls verify and update me. check the vs code part also...i have created a subfolder under monopoly folder called code for the source code for this project.	Checked the installer report: Temurin 25.0.5 LTS, javac, JAVA_HOME, VS Code Java extensions, Git, compile test all passed. Found the extra JDK 17; pinned VS Code to JDK 25; added Check-Java.bat.
01.07	16:45	**Requirements**	advise me on the group's three twists, i didnt get it...what i need to do for this?	Explained the pool of eight twists and that the instructor assigns three per group on Blackboard; asked CS to confirm the group number and twists.
01.08	16:47	**Requirements**	i want to use only simple ones for the twists without disrupting too much and easy to implement. so suggest me 3 twists	Challenged choosing twists (they are assigned); if a choice is allowed, recommended T1 Reverse, T3 Expansion, T7 Recall.
01.09	16:53	**Requirements**	Ok for now lets not build the twists, will enquire about this what twist is required for this group and we will enhance this later. Let me know about the bonus mark and what are we planning to do for that?	Recorded the decision to defer twists; explained the bonus rules (0.25 each, unique, judged in demo order) and proposed five candidate features.
01.10	16:58	**Requirements**	What about adding a feature to change the monopoly game language as options for this bonus. there will be toggle to select language like english, french, arabic. whats your view on this? since the mark is only .25 should be touch complex parts	Assessed the language toggle: little data-structure content, ResourceBundle/HashMap banned, Arabic console risk; suggested an own recursive binary-search lookup.

No.	Time (GST)	Phase	Prompt by CS (verbatim)	AI output and outcome
01.11	17:02	**Requirements**	Ok lets use option B4 and also language toggle option for bonus. keep this in memory	Recorded the bonus choice: B4 pointer visualiser and a language toggle (English, French, Arabic); saved the decisions to project memory.
01.12	17:04	**Design**	ok now lets move to the next step the core design and build the uml class diagram, plus board and card files	Class model to the Section 5 signatures; UML diagram; own board.txt (24 squares) and cards.txt (20 cards) checked by script; design notes D1-D13.
01.13	18:16	**Core game**	ok now, based on this design build the monopoly game application using java and requirements mentioned in the context document	Built the core game (24 Java files) with the 8 required recursive methods and no loops; 44 tests incl. every Section 8.2 scenario; constraint checker; build/run/test scripts; README. Found and fixed 3 AI issues.
01.14	18:35	**Core game**	i dont understand why it is a command like game, i thought there will be visualisation and it will be a console game like the gone shown in the image initially.	Acknowledged the miss (the prototype showed a board); added BoardView: ring of squares, owners, tokens, player table, turn queue, events in plain words. --text keeps the Appendix C log mode. 50 tests.
01.15	18:45	**Core game**	the console game is running in black and white mode, like shown in the prototype image, can you pls make the game in color mode, so that it looks atheistically better. choose different color for different players and the use that color for the property they own. make these enhancements	Switched on colour in the classic Windows console; each player has a colour that tints the squares they own; changed the palette after a green-on-green clash.
01.16	18:49	**GitHub**	now i need this project to be synced with the github branch created for the person in the project group. Pls sync the changes to https://github.com/cb-beep-netizen/CSC301-Group4-Monopoly/tree/CS-Monopoly	GitHub not linked to the session; read the public repository; found teammates' files and twists T1/T5/T8 in a teammate's log; wrote sync-to-github.bat (never deletes teammates' files, keeps private files out).
01.17	18:54	**GitHub**	check the error screenshot, the github signin page did not open...pls fix this (Attachment: screenshot of sync-to-github.bat ending "SYNC FAILED - see sync-log.txt" after step [5/5] Pushing to GitHub.)	Diagnosed HTTP 403: the saved GitHub account had no write access. The script now explains the fixes and can switch account.
01.18	18:57	**GitHub**	yes, i want to keep my email private, pls make the necessary changes so that my email is not exposed	Sync script signs commits with the GitHub no-reply address and re-signs unpushed commits; rehearsed on a copy of the branch.
01.19	not recorded	**GitHub**	Pls connect to the github repo branch https://github.com/cb-beep-netizen/CSC301-Group4-Monopoly/tree/CS-Monopoly and sync the folder C:\claude\monopoly to github branch mentioned in the link	(Separate Claude session) Checked the link; dry-run push failed with 403; flagged the e-mail in the unpushed commit and a changed VS Code AI setting.
01.20	not recorded	**GitHub**	ok login to a different github account which has access. i have already logged in this account using my chrome browser	(Separate Claude session) Signed in with the team account; dry-run push succeeded; re-signing was blocked by a permission check.
01.21	19:12	**GitHub**	yes log this request in AI-usage-log.txt file. try to connec to repo using [e-mail removed from this log: the cb-beep-netizen account] account and copy the monopoly project to the branch https://github.com/cb-beep-netizen/CSC301-Group4-Monopoly/tree/CS-Monopoly	(Separate Claude session) Logged Prompts 01.19-01.21 (e-mail removed); pushed the project to CS-Monopoly with a no-reply address (commits aabd08f, b2578b0).
01.22	19:24	**Enhancements**	I need some enhancements to the program. at the end of the game, Pls show the ranking of the players, how much is their networth and list the properties owned by them. this should come when the game is quit. also in the console centre also currently where the ranking, networth is show, pls add the properties owned by them also.	Centre panel lists each player's properties; quit and game-over screens show the ranking, net worth (cash + property) and full property lists. Q writes no END line (Appendix C). 53 tests.
01.23	19:32	**Reporting**	make a pdf report based on the AI-Usage-log log file. the report must be titled appropriately using the context for the project name, sub heading as AI usage log for user CS (Caron Sylvia). The report should initially show a summary section with a small write up on AI usage and a pictorial timeline of AI prompts with functionalities build. Then the detailed report section should show the tabular view of the AI log usage. as required in the context word doc uploaded initially. keep this pdf report also in the monopoly folder	Generated this report from AI-Usage-log.txt.

2.3 AI reflection - candidate cases (brief Section 7.1)

Cases where the AI's output was wrong, incomplete or broke a constraint, recorded as they happened. The brief asks for at least three, with how each was detected and fixed.

Case	Prompt	What the AI produced	How it was detected	How it was fixed
1	01.03, found at 01.06	Stated that no JDK was installed, after checking only standard folders on drive C:.	The installer's "where java" output also listed F:\Android\jdk-17.	VS Code pinned to Temurin 25; Check-Java.bat added; skill now checks other drives.
2	01.02	A prototype screen whose cash and net-worth figures were invented.	Reasoning (flagged by the AI itself).	Marked as illustrative; real values come from seeded test games.
3	01.13	A board whose rents never bankrupt anyone (0 bankruptcies in 300 games of up to 3,000 turns).	Simulation of seeded games.	Rents x2/x3/x4 tested (still 0 in 60 turns); brief-like rents kept; bankruptcy shown with scripted dice; documented.
4	01.13	Test code using java.util.Arrays (not allowed) and a constraint checker with false positives.	Reviewing imports; reasoning about the checker output.	Own list() helper; word-boundary patterns in the checker.
5	01.13, found at 01.14	A game that printed only log lines, although the AI's own prototype showed a console board.	CS played the game and compared it with the prototype (wrong result while playing).	BoardView added as the default screen; --text keeps the Appendix C log mode.

Still to be written by the group: the 1-2 page reflection in the group's own words, and the closing paragraph on which parts the group designed or rewrote itself, and why (Section 7.1).

2.4 Files produced, traced to prompts

Brief Section 7.3: code that appears in no log, and that no member can explain, is treated as unlogged AI use. Each AI-produced file also names its prompt in its header.

File	Prompt(s)	Purpose
AI-Usage-log.txt	01.01 (every prompt)	Audit trail required by brief Section 7
Prototype_Console_Mockup.png	01.02	Console UI prototype
setup\Setup-Environment.bat/.ps1, Check-Java.bat	01.05, 01.06	JDK 25 installer and checks
skills\csc301-java-environment-setup\SKILL.md	01.05	Reusable set-up procedure
.vscode\settings.json	01.05, 01.06, 01.13	Workspace settings (Java paths; AI features)
design\UML_Class_Diagram.*, Core-Design-Notes.txt	01.12-01.22	Class diagram v3, design decisions D1-D21
code\data\board.txt, cards.txt	01.12	Own board (24 squares) and 20 cards
code*.java (25 files)	01.13-01.15, 01.22	Game source incl. BoardView
tests\GameTests.java, check_constraints.py	01.13-01.22	53 tests; constraint check
build.bat, run.bat, run-tests.bat, README.txt	01.13-01.22	Build, run, test; README deliverable
sync-to-github.bat	01.16-01.18	GitHub sync with private e-mail
CSC301_Group4_Monopoly_AI_Usage_Report_CS.pdf	01.23	This report